Blessed G/Michael Catholic Secondary School

Semester 1 IT Mid1 Examination for Grade10

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade10 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** Which component of the Central Processing Unit (CPU) performs mathematical calculations and logical comparisons?

A. Control Unit (CU) **B.** Arithmetic Logic Unit (ALU)

C. Registers **D.** Cache Memory

____ **2.** Which type of network topology connects all nodes to a central hub or switch?

A. Bus topology **B.** Star topology **C.** Ring topology **D.** Mesh topology

____ **3.** An IPv4 address consists of how many bits?

A. 16 bits **B.** 32 bits **C.** 64 bits **D.** 128 bits

____ **4.** Which of the following is volatile memory that loses its contents when power is turned off?

A. ROM **B.** RAM

C. Hard Disk Drive **D.** Flash Memory

____ **5.** Which optical storage media typically holds up to 700 MB of digital data?

A. CD-ROM **B.** DVD-ROM **C.** Blu-ray Disc **D.** Magnetic Tape

____ **6.** Which computer component synchronizes all internal operations using periodic electrical clock pulses?

A. System Clock **B.** Hard Disk Drive

C. Graphics Processing Unit **D.** Network Switch

____ **7.** How many bytes are in exactly one Kilobyte (KB) in binary computing (IEC standard)?

A. 1,024 bytes **B.** 1,000 bytes **C.** 512 bytes **D.** 2,048 bytes

____ **8.** Which type of computer software coordinates hardware operations and provides a platform for application programs?

A. System Software (Operating System) **B.** Application Software

C. Utility Software **D.** Firmware only

____ **9.** Which network covers a small localized geographic area such as a school laboratory or office building?

A. Local Area Network (LAN) **B.** Wide Area Network (WAN)

C. Metropolitan Area Network (MAN) **D.** Personal Area Network (PAN)

____ **10.** What is the function of the BIOS (Basic Input/Output System) chip during computer startup?

A. Executes the Power-On Self-Test (POST) and loads the operating system kernel into RAM
B. Connects to the Wi-Fi router automatically
C. Formats infected hard drives
D. Runs spreadsheet calculations

____ **11.** Which transmission medium uses pulses of light transmitted through glass cores to transfer data at the speed of light?

A. Fiber optic cable **B.** Coaxial cable

C. Shielded twisted pair **D.** Copper telephone wire

____ **12.** What is phishing in information security?

A. A fraudulent attempt to steal sensitive passwords and financial data by impersonating trustworthy entities in emails or websites
B. Searching for files in Windows Explorer
C. Catching optical cable signals with radio
D. Backing up database tables to flash drives

____ **13.** Which printer technology sprays microscopic droplets of liquid ink onto paper to produce high-resolution color documents?

A. Inkjet printer
B. Laser printer
C. Dot matrix printer
D. Thermal printer

____ **14.** What is open-source software?

A. Software whose source code is freely available for inspection, modification, and redistribution (e.g. Linux)
B. Software that requires commercial paid licenses only
C. Software that only works without an internet connection
D. Software that expires after 30 days

____ **15.** What does the acronym GUI stand for in operating systems?

A. Graphical User Interface
B. General Universal Instruction
C. Global Unit Interconnect
D. Geographic User Index

____ **16.** Which port interface is the modern universal standard for connecting peripherals, keyboards, mice, and flash drives?

A. USB (Universal Serial Bus)
B. VGA port
C. Parallel LPT port
D. Serial RS-232 port

____ **17.** Which of the following is an example of an input device?

A. Scanner and Optical Barcode Reader
B. LCD Monitor
C. Laser Printer
D. Audio Speakers

____ **18.** What is the primary function of an Uninterruptible Power Supply (UPS)?

A. Provides immediate battery backup power during mains power failure to prevent sudden data loss
B. Increases computer CPU processing speed
C. Supplies high-speed internet data
D. Cools down motherboard components

____ **19.** In computer networking, which device modulates digital computer signals into analog frequencies for telephone line transmission?

A. Modem (Modulator-Demodulator)
B. Hub
C. Repeater
D. Bridge

____ **20.** What is cloud storage (such as Google Drive or OneDrive)?

A. Saving digital files on remote online servers accessible over the internet rather than local hard drives
B. Storing data inside storm clouds with radio waves
C. Backing up files to DVD discs
D. Printing files to paper archives